

Writing Workbook

From first sentence to complete draft: Semester 1, Tutorials 2-8

Name	
Tutor / group	
My research question (fill in on 29 Oct)	

HOW THIS WORKBOOK WORKS

Every tutorial from 8 October produces one **piece** of your report. The tasks between tutorials are not extra work: they **are** the report, written in small pieces.

By **Thursday 3 December** you will have a complete first draft. December and January are for editing, and you submit your draft on **Friday 5 February 2027**.

Bring this workbook (printed or on screen) to every tutorial. Fill it in by hand or type into it; either is fine.

Page references are to Schultz, D. M. (2009) *Eloquent Science*, American Meteorological Society, the course writing text.

1. Key dates

Date	What happens	What you bring / produce
Thu 1 Oct	Tutorial 1: launch the notebook	Notebook running; two patterns noted
Thu 8 Oct	Tutorial 2: who reads your report, and how?	'Explain it to a friend' (150 words)
Thu 15 Oct	Tutorial 3: hook, problem, purpose (Introduction)	Purpose statement, hook, working title
Thu 22 Oct	No tutorial (reading week)	Tasks 3A–3C
Thu 29 Oct	Tutorial 4: storyboard and outline	One committed question ; storyboard
Thu 5 Nov	No tutorial	Figures; Methods draft (4B)
Thu 12 Nov	Tutorial 5: figures as evidence	Peer-checked Methods; captions
Thu 19 Nov	Tutorial 6: paragraphs that argue	Results draft, marked up
Thu 26 Nov	Tutorial 7: Discussion and Conclusion	Introduction draft; Discussion plan
Thu 3 Dec	Tutorial 8: draft complete	Complete first draft ; revision plan
Dec – Jan	Rest, then three editing passes	See your revision plan (Tutorial 8)
Fri 5 Feb	Draft submitted (formative: no marks; no DASS extensions)	Your tutor confirms the exact method
w/c 8 or 15 Feb	Swap drafts: peer feedback	Feedback for a classmate
w/c 1 Mar	Group presentation in your tutorial	
Fri 12 Mar, 12:00	Final report submitted (DASS extensions apply)	Max 2,000 words
Wed 7 Apr	Marks and feedback released	

THE REPORT

Individual report, **maximum 2,000 words**, 100% of the course mark. One focused question, answered with about **3–5 figures**.

2. AI Minimal: what it means for you

THE POLICY FOR THIS REPORT

“AI may be used only for limited presentation-related support, such as grammar, fluency, spelling, or formatting. Students' ideas, arguments, and conclusions must remain their own.”

Allowed	Not allowed
▪ Grammar, spelling and punctuation checks on text you wrote ▪ Fluency suggestions you accept or reject yourself ▪ Formatting (headings, references layout, figure placement)	▪ Choosing or refining your research question ▪ Generating structure, arguments, interpretations or conclusions ▪ Summarising papers or explaining your results to you ▪ Drafting or rewriting sentences or paragraphs

Protect yourself (and work like a scientist)

- Save **dated versions**: report_v01_2026-11-12.docx, report_v02_2026-11-19.docx ... (Schultz keeps numbered versions of everything, §6.2).
- Keep your **writing log** (Section 4). Together with your versions, it shows how your work developed.
- Unsure whether something is allowed? Ask your tutor **before** you do it.

3. Your 2,000-word budget

Section	Words	Paragraphs	Its job
Introduction	~350	~3	Context → problem ('however') → purpose statement
Methods	~300	2–3	Data, period, region, ONI and thresholds, anomalies, weighting, statistics
Results	~650	4–6	One paragraph per claim/figure: what you found, quantified
Discussion	~450	3–4	Mechanism, comparison, exceptions, limitations, back to the question
Conclusion	~150	1–2	Answer the question; 2–3 findings; no new material
Spare	~100		For whichever section needs it

This budget assumes figure captions, the reference list and the title are outside the 2,000 words. Your tutor will confirm what counts.

4. Writing log (Task 2C, all term)

After **every** notebook session, write three short lines. It takes five minutes, and it becomes the raw material for your Results and Discussion.

Date	What I ran	What I noticed (with numbers)	What puzzled me

What are we investigating?

Goal: Get the notebook running and see your first cloud figures.

THE PROJECT IN ONE LINE

Answer **one focused question** about ENSO and cloud, using about **3–5 figures**, in a report of at most **2,000 words**. The notebook is a toolbox, not a checklist.

TASK 1B Two patterns, written properly • 15 min • due 8 Oct

	Where (region)	What (which cloud variable, what it does)	When (season / ENSO phase)	Why might it happen?
1				
2				

Task	Done?
1A Run notebook sections 0–3; save figures outside Binder	☐
1C Read handbook §1–3; skim §9 (marking)	☐

Who reads your report, and how?

Goal: Know what your report has to do for its reader, and start writing about your project this week.

The four things your marker must believe

Proposition	Your marker asks...	Mostly shown in...
Importance	Is this question worth asking?	Introduction
Success	Does the report actually answer it?	Results + Conclusion
Value	Does every figure and sentence earn its place?	The whole report
Competence	Are the methods sound and the limitations honest?	Methods + Discussion

Adapted from the four propositions of the Research Funding Toolkit.

Methods today, one sentence each

After each methods explanation, write the sentence you would put in your Methods. Then check it against the tutor's version.

Idea	My Methods sentence
ERA5 cloud cover (which variable, which period)	
Area weighting	
Climatology and anomalies (which baseline)	

THREE FACTS WORTH REMEMBERING

TCC ≠ LCC + MCC + HCC: cloud layers overlap, so the total seen from above is less than the sum.

Changes in cloud cover are in **percentage points (pp)**: 40% → 44% is +4 pp (a 10% relative change).

ERA5 cloud comes from the model's cloud scheme, constrained only indirectly by observations: trust the **sign** more than the exact **size**.

In class: explain it to a friend (150 words, by hand, 7 minutes)

Explain to a friend who studies something else: **(a)** what a cloud-cover **anomaly** is, and **(b)** why ENSO might change cloud somewhere.

Write here. Partner: underline the clearest sentence; circle the first place you had to reread.

TASK 2A Explain it to a friend (final) · 20 min · due 15 Oct

- Redraft to ≤ 200 words. Fix the circled sentence.
- Add one sentence on **where** you would look for an ENSO cloud signal, and why.

Final version (word count: ____)

TASK 2B Reverse-outline a published paper · 45 min · due 15 Oct

Find one paper on ENSO and clouds (Google Scholar or the library: try “ENSO cloud cover reanalysis”, “El Niño low cloud southeast Pacific”, “ENSO high cloud tropical Pacific”). For each paragraph of its **Introduction**, record its job (context / problem / purpose / scope / organisation) and its point.

Reference: _____

¶	Job	Its point, in ≤ 12 words
1		
2		
3		
4		
5		

¶	Job	Its point, in ≤ 12 words
6		

Where is the **'however'** (the problem)? _____ Where is the **purpose** statement?

TASK 2D Optional: follow the 2026 El Niño live · 10 min · due 15 Oct

NOAA expects a very strong El Niño this winter (as of September 2026). Look up this week's **Niño 3.4** value on NOAA CPC's ENSO page and write one sentence on what it might mean for cloud over the central Pacific and the Maritime Continent. *Remember: this event is not in your dataset (it ends in 2025), so it is context, not evidence.*

TASK 2C Set up: writing log + reference manager · 20 min · due 15 Oct

- **Writing log:** use Section 4 of this workbook after every notebook session.
- **Reference manager:** install **Zotero** (free, zotero.org, plus its browser connector and Word/Google Docs plugin) or **EndNote** (licensed by the University; free via the software download service; Harvard Manchester style; Library workshops). Avoid starting on Mendeley Desktop (legacy, no further updates).
- Save your 2B paper into it, and choose **one author-year citation style**.

Literature toolkit: searching Web of Science

Step	How
Sign in	webofscience.com → sign in through the University (institutional login)
Search	Advanced Search , Core Collection selected. Fielded Search tab: one idea per Topic row, joined by AND, no TS= tags. Query Builder tab: paste a tagged query straight into the Query Preview box. Field tags (Query Builder only): TS= topic, TI= title, AU= author, PY= year. Phrases in "quotes"; AND / OR / NOT; cloud* matches cloud, clouds, cloudiness; NEAR/3 for words close together
Refine	Document Types → Review Article; Categories → Meteorology & Atmospheric Sciences; publication years
Sort	Citations (highest first) for the classics; Date for the newest
Chase	Cited References (backward) · Times Cited (forward) · Related Records (similar)
Combine	Search History: #1 AND #2
Save	Zotero Connector (one result or a whole page), or Export → RIS / EndNote

TRY THIS SEARCH, THEN ADAPT IT TO YOUR QUESTION

TS=("El Nino" OR ENSO OR "Southern Oscillation") AND TS=("cloud cover" OR "cloud fraction" OR "cloud amount") AND TS=(reanalysis OR ERA5 OR satellite)

Paste it into **Advanced Search** → **Query Builder**, straight into the **Query Preview** box (not the term box above it, which wraps it in ALL=(...)). Or, in **Fielded Search**, use three Topic rows: "El Nino" OR ENSO OR "Southern Oscillation" · AND "cloud cover" OR "cloud fraction" OR "cloud amount" · AND 'reanalysis OR ERA5 OR satellite'. Type 'El Nino' without the tilde. Refine to Review Article and sort by Citations. Then

replace the cloud terms with your own cloud type or region.

Scholar or Web of Science? Scholar to explore broadly and get full texts fast; Web of Science for a precise search of peer-reviewed journals, with reliable citation counts and filters.

Zotero in Word: set up once, then three rules

- Install Zotero (desktop) and the Zotero Connector from [zotero.org](https://www.zotero.org). The Word plugin installs itself; restart Word and a **Zotero tab** appears (Windows and Mac).
- **Add/Edit Citation** inserts a citation; **Add/Edit Bibliography** builds the reference list; **Document Preferences** sets the style. Google Docs: use the Zotero menu the Connector adds.
- **Word for the web (OneDrive in a browser) is not supported:** open the file in the desktop Word app to cite.

Rule	
1 Never hand-edit	Zotero overwrites manual changes to citations or the reference list. Fix the entry in your Zotero library.
2 One style, chosen once	Pick an author-year style in Document Preferences and keep it.
3 Share a PDF	For the February peer swap, send a PDF so citations show correctly for someone without Zotero.

Using EndNote instead? Its 'Cite While You Write' tab in Word does the same job, and the Library supplies a Harvard Manchester style.

WHERE TO SEARCH

Google Scholar: Settings → Library links → University of Manchester, so full texts open via your university login.

Web of Science (formerly 'Web of Knowledge'; [webofscience.com](https://www.webofscience.com), sign in through the University): start from one good paper; go **backward** (its references) and **forward** ('Times cited'); filter for review articles.

Library Search for books and full text. Core journals: *Journal of Climate*, *J. Atmos. Sci.*, *Geophys. Res. Lett.*, *Q. J. R. Meteorol. Soc.*

NOAA Climate.gov is excellent **background**; scientific claims need **peer-reviewed** sources.

AI Minimal: don't use AI tools to find, choose or summarise papers for you.

NOTEBOOK STRAND AND READING (BEFORE 15 OCT)

Notebook: run sections 1–3 (seasonal cycle; global ENSO maps; global-mean anomalies). Analysis slides 02 and 03.

Read: Schultz §4.1–4.2 (parts of a paper; nonlinear reading) and ch. 7 (accessible scientific writing).

Hook, problem, purpose: the Introduction

Goal: Draft the opening and the purpose statement for a question you might investigate.

The three moves of an Introduction (Schultz §4.6)

Move	What it does	Pitfall
1 Context	Common ground the reader already accepts	The 'resounding banality': a true, obvious opening that does no work
2 Problem	The hook: a contrast, puzzle, gap or stake ('However, ...')	Claiming 'no one has studied this': you can't know that
3 Response	"The purpose of this report is to ...", plus its scope	A topic ('to look at ENSO and clouds') instead of a purpose

Methods: thresholds and composites

THE NUMBERS (ONI, JANUARY 2005 – JULY 2025, $\pm 0.5^\circ\text{C}$, MONTH BY MONTH)

54 El Niño months • 82 La Niña • 111 neutral. At $\pm 1.0^\circ\text{C}$: 24 El Niño, 25 La Niña.

The 2015–16 event alone supplies 14 of the 54 El Niño months.

NOAA's official episodes need $\pm 0.5^\circ\text{C}$ for ≥ 5 consecutive overlapping seasons; the notebook classes each month on its own. Say which you used.

Composite = mean anomaly over El Niño months – mean anomaly over La Niña months (units: pp). It's a difference between two states, not a correlation.

In class: predict before you plot (El Niño – La Niña, total cloud)

Region	My prediction (+ / - / 0)	Physical reason	Actual
4 Niño 4 (central Pacific)			
5 Maritime Continent			
6 SE Pacific stratocumulus			

Any surprise is a candidate '**however**' for your Introduction.

In class: from a pattern to a purpose

Pattern 1 I noticed (where, which cloud, what sign):

Nutshell: The purpose of this report is to ...

verb (quantify / compare / test whether) • what • where/when • how

Hook (one sentence of common ground, then “However, ...”):

Working title (informative · accurate · clear · concise · attention-commanding):

Pattern 2 (optional): nutshell

TASK 3A Opening paragraph(s), ~250 words · 45 min · due 29 Oct

- For your **leading** candidate question: context → ‘however’ → purpose statement.
- Check sentence 1: would anyone disagree with it, or learn anything from it? If not, rewrite it.

Draft (word count: ____)

TASK 3B Literature synthesis paragraph · 75 min · due 29 Oct

Find **three relevant sources** (at least two peer-reviewed). Fill in the table, then write **one ~150-word paragraph** that synthesises them around a single idea. The **idea** is the subject of each sentence; the citations go in brackets.

Full reference	Key finding (with a number if possible)	How it relates to my question

Full reference	Key finding (with a number if possible)	How it relates to my question

GROCERY LIST VS SYNTHESIS (SCHULTZ §4.7)

× “Smith (2001) studied low clouds. Jones (2005) looked at ENSO. Lee (2012) found that SST affects clouds.”

✓ “Subtropical low cloud is closely tied to lower-tropospheric stability (Smith 2001; Lee 2012), so ENSO-driven changes in sea-surface temperature could alter it ...” (*placeholder names*)

Always cite your data: ERA5: Hersbach, H. et al. (2020) The ERA5 global reanalysis. *Q. J. R. Meteorol. Soc.*, 146, 1999–2049. ONI: NOAA Climate Prediction Center (state the version/date you downloaded).

Synthesis paragraph (word count: ____)

TASK 3C Three candidate questions · 30 min · due 29 Oct

	Question	I find this interesting because...	The figure that would answer it
1			
2			

	Question	I find this interesting because...	The figure that would answer it
3			

NOTEBOOK STRAND AND READING (BEFORE 29 OCT)

Notebook: sections 4–5 (regional analysis; ONI vs regional anomaly), plus 6 if timing interests you. Explore low/medium/high cloud in your candidate regions. Optional: methods_demos.ipynb §1–2. Analysis slides 04.

Read: Schultz §3.1–3.2 (titles), §4.6–4.7 (introduction; literature synthesis), §12.1–12.3 (citing).

Plan the story before you write it

Goal: Leave with one question, a figure storyboard and a topic-sentence outline.

Is my question good enough?

Test	Yes?	Notes
Specific: names a cloud type, region, season or timing		
Answerable with ERA5 + ONI and the notebook's tools		
Interesting: there is a real 'however'		
Bounded: 3–5 figures, ~650 words of Results		
Physical: there is a mechanism I can discuss		

Methods: which tool answers my question?

Tool	Question it answers	Units	Watch out for
Composite	How different are El Niño and La Niña months on average?	pp	Few events; threshold choice; hides asymmetry unless split by phase
Correlation r	How consistently does cloud follow the ONI?	none (–1 to +1)	Says nothing about size; straight lines only; needs n
Regression slope	How much does cloud change per °C of ONI?	pp per °C	Straight line; pulled by extremes (2015–16)

$R^2 = r^2$ is the fraction of variance a straight line explains ($r = 0.5 \rightarrow 25\%$). A regional mean can **cancel** if the box straddles positive and negative anomalies: choose boxes on one coherent signal, and test by shifting the box 5°.

My region(s) and bounds: _____

Tool(s) I will use, and why: _____

My committed question:

Storyboard: one claim under every figure

Sketch each figure (rough is fine), give it a role (context / test / quantify / qualify / compare), and write the claim it proves. Use [] for numbers you don't have yet.

Fig.	Sketch + what is plotted	Role	This figure shows that...
1			
2			
3			
4			
5			

Partner check: do the claims, read in order, answer the question? Which figure could be removed?

TASK 4A Make the storyboard figures · notebook time · due 12 Nov

- Adapt the notebook analyses to **your** region / cloud level / season. Save draft figures **outside Binder**. Note key numbers in your writing log.

TASK 4B Methods draft, 250-300 words · 60 min · due 12 Nov (bring for peer check)

Write the science, not the code. Use this checklist:

- ☐ Dataset: ERA5 monthly means; which **cloud variable(s)** (TCC / LCC / MCC / HCC), cited
- ☐ **Analysis period** (first and last month)
- ☐ **Region(s)**: name (tag-phrase) and latitude/longitude bounds
- ☐ ENSO index: **ONI**, what it is, source; El Niño / La Niña **thresholds** and how months were selected
- ☐ **Anomalies**: relative to which monthly climatology (which years)
- ☐ **Area weighting** (cos latitude) for regional means
- ☐ **Statistics** you actually use: composite difference, correlation (r), regression slope, lag, any significance test, with **n** (and **n_eff** if you test significance)
- ☐ No Python, no cell numbers, no interpretation (that's Discussion)

Methods draft (word count: ____)

TASK 4C Full topic-sentence outline · 40 min · due 12 Nov

One sentence per paragraph, for the whole report. Read only these sentences, in order: they should tell your whole story.

Section (budget)	¶	Topic sentence
Introduction (350)		
Methods (300)		
Results (650)		
Discussion (450)		

Section (budget)	¶	Topic sentence
Conclusion (150)		

NOTEBOOK STRAND AND READING (BEFORE 12 NOV)

Notebook: analysis slides 04 and 05; optional section 6 if your question is about timing.

Read: Schultz ch. 6 (brainstorm, outline, first draft), §4.8 (data and methods), §18.3 (picking thresholds).

Make every figure earn its place

Goal: Leave with report-ready captions and a Methods section someone else could repeat.

In class: could I repeat your study? (peer check)

Reviewer: list every decision you would have to **guess** to redo the analysis, and anything that is code rather than science.

Reviewer's name: _____

Guess list

Caption check (Schultz §11.10 + handbook §8)

- ☐ Begins with a phrase capturing the figure's **essence**, different from every other caption
- ☐ **What** is plotted (raw field, anomaly, difference, composite, correlation)
- ☐ **Units**
- ☐ **Period** and **region**
- ☐ Any **threshold or definition** (e.g. how El Niño months were chosen)
- ☐ Every **panel** and symbol explained; the **sign convention** is clear
- ☐ Diverging colour map **centred on zero** with symmetric limits; panels you compare share the **same** limits; no rainbow
- ☐ **Attribution:** "Figure produced by the author using ERA5 (ECMWF/Copernicus Climate Change Service) and ONI (NOAA Climate Prediction Center) data."

In class: caption clinic

Fig.	Full caption	The result sentence that cites it: science first, "(Fig. n)" last
1		
2		

Fig.	Full caption	The result sentence that cites it: science first, "(Fig. n)" last
3		
4		
5		

TASK 5A Revise Methods · 30 min · due 19 Nov

- Fix every item on the guess list; remove code and interpretation; stay within ~300 words.

TASK 5B Results draft, ~550-650 words · 90 min · due 19 Nov

One paragraph per storyboard claim. Plan each with the pattern **Claim → Evidence → Interpretation (light) → Qualification**:

¶	Claim (topic sentence)	Evidence: numbers + figure	Qualification / exception
1			
2			
3			
4			
5			

Keep a 'Discussion parking lot' at the end of your draft for explanations that occur to you. Results say **what**; the Discussion says **why**.

TASK 5C Final captions · 20 min · due 19 Nov

- All figures, one consistent format, numbered in the order you cite them.

TASK 5D Calculate a p-value yourself · 20 min · due 19 Nov

WHY R IS NOT ENOUGH: COINS IN A BUCKET

Pull the coins 1p, 2p, 5p, 10p, 20p, 50p, £1 out of a bucket at random. The chance they come out in perfect order ($r = 1$ between draw order and value) is $1/n!$: **1 in 6** for 3 coins, **1 in 120** for 5, **1 in 5,040** for 7. The same perfect r can be likely or very unlikely to be luck. The p-value combines **r** and **n** .

$t = r \sqrt{(n - 2) / (1 - r^2)}$, with $n - 2$ degrees of freedom; p is the two-tailed probability of $|t|$ at least this large if the true correlation were zero. Use methods_demos.ipynb §4, a calculator with a t-table, or `stats.t.cdf` in Python.

	r	n used	t	p	Significant at 0.05?
Using n (months, from the \$5 panel)					
Using n_eff (from the \$6 table)					

Which would you report, and why? _____

n	5	10	30	60	100	247
r needed for $p < 0.05$	0.88	0.63	0.36	0.25	0.20	0.12

$n_{\text{eff}} = n (1 - r_1^* r_{1y}) / (1 + r_1^* r_{1y})$, where r_1 is each series' lag-1 autocorrelation. The ONI's r_1 is 0.97, so 247 months behave like roughly 30–135 independent ones, depending on the cloud series.

Optional: is one year different from another? (notebook §3b)

Set YEAR_A and YEAR_B **before** looking at the plot, for a physical reason (e.g. a strong El Niño year vs a La Niña year).

Years compared	Difference (pp)	p (naïve)	p (adjusted)	With DETREND = True

Remember: 20 complete years make 190 possible pairs, so about 10 will look 'significant' at $p < 0.05$ by chance alone.

NOTEBOOK STRAND AND READING (BEFORE 19 NOV)

Notebook: finalise 3–5 figures (labels, units, centred colour scales); save them outside Binder. Work through methods_demos.ipynb §3–6, and run the shift test (§7) for your region. Optional: §3b to compare years; a ± 1.0 °C sensitivity test.

Read: Schultz §11.10–11.14 (captions; discussing and citing figures) and §10.1 (concision).

Paragraphs that argue

Goal: Turn your Results draft into paragraphs that argue, not describe.

The order of an argument (Schultz §7.3)

data → results → interpretation → inference → speculation. Assert the **finding** first (topic sentence), then the evidence. **Never lead with speculation**, and flag it when you use it (“We speculate that...”).

Methods: reporting a correlation in full

TEMPLATE (ILLUSTRATIVE NUMBERS)

“Monthly high-cloud anomalies over [region] are negatively correlated with the ONI ($r = -0.45$, **95% range** -0.63 to -0.22 ; $n = 247$ months, $n_{\text{eff}} \approx 60$; $p < 0.001$; **January 2005 – July 2025**), a decrease of about **2.1 pp per °C** of ONI.”

Is my response symmetric? (from the notebook's phase table, sections 3–4)

Region / variable	El Niño mean (pp)	Neutral mean (pp)	La Niña mean (pp)	Symmetric? One-sided?

In class: paragraph surgery

- ☐ **Highlight the first sentence** of every Results paragraph. Read only those: do they answer the question, in order?
- ☐ Label each sentence **C / E / I / Q**. Move deep interpretation to the Discussion parking lot.
- ☐ **Circle every number**. Units? Baseline? Period or n ? Percent vs **percentage points**? Sensible precision ($r = -0.52$, not -0.5183)?
- ☐ No sentence just describes the plot (“Figure 3 shows...”, “It can be seen that...”).
- ☐ Find the weakest link between two sentences; repair it with repetition, a transition, or reordering (old → new information).
- ☐ Partner marked where they got lost in one paragraph.

Fixes I need to make

TASK 6A Revise Results • 45 min • due 26 Nov

- Apply the surgery to every paragraph; stay within ~650 words.

TASK 6B Introduction, full draft ~350 words · 60 min · due 26 Nov

- Rewrite Task 3A now that you know your results. Fold in your 3B synthesis paragraph. End with the purpose statement (and hypothesis if useful).
- Use exactly the same tag-phrases as in Methods and Results.

TASK 6C Discussion brainstorm · 30 min · due 26 Nov

	Possible physical explanations for my main result
1	
2	
3	

	Limitations that genuinely matter for my answer (and their likely effect)
1	
2	

Devil's advocate: the strongest objection a critic could raise is...

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NOTEBOOK STRAND AND READING (BEFORE 26 NOV)

Notebook: only what your Results need (fill any [] numbers). No new directions: your question is fixed.

Read: Schultz ch. 8 (paragraphs), §7.3 (logical arguments), §9.1 (active vs passive voice).

What does it mean?

Goal: Leave with a Discussion plan and a Conclusion that answers your question.

In class: plan the Discussion (choose 3-4 moves)

Move	Use it?	Topic sentence for this paragraph
EXPLAIN (compulsory): the physical mechanism	✓	
COMPARE with published work		
EXCEPTIONS: events, regions or seasons that don't fit		
ALTERNATIVES: another explanation for the same pattern		
LIMITATIONS that affect the answer		
BACK TO THE QUESTION (compulsory)	✓	

Methods: from correlation to mechanism

Cloud is a symptom. Set VARIABLE to **skt** (skin temperature $\approx$ SST over the ocean), **tcwv** (water vapour) or **tclw** / **tcw** (cloud liquid / ice) and rerun §2 and §5. Does the supporting field tell the same story?

Variable	Sign of El Niño – La Niña over my region	Consistent with my mechanism?
skt		
tcwv		

Lags: a peak at +3 months is consistent with a delayed response but doesn't prove cause. Peaks are broad because both series have long memory. Write 'is associated with' or 'is consistent with' unless you argue the mechanism.

One strong limitation

What is limited	
How much (number of events, years...)	

Likely effect on my answer	
A check that would test it	

The mirror test

Last sentence(s) of my Introduction (purpose)	First sentence of my Conclusion (answer)

- ☐ Same question, not more, not less
- ☐ Same tag-phrases (regions, variables, index)
- ☐ A reader of only these two boxes would know my question **and** my answer

TASK 7A Discussion draft, ~450 words · 75 min · due 3 Dec

- Your 3–4 moves, one paragraph each. Mechanism, honest limitation, back to the question. Flag speculation as speculation.

TASK 7B Conclusion, 100–150 words · 25 min · due 3 Dec

- Sentence 1 answers the question. Two or three findings with their key numbers. No new results, figures or citations. End with something specific, not “more research is needed”.

TASK 7C Assemble the complete first draft · 60 min · due 3 Dec (bring printed or as PDF)

- ☐ Title (revisited: is it accurate now?)
- ☐ Introduction · Methods · Results · Discussion · Conclusion
- ☐ 3–5 figures, each with a full caption and attribution
- ☐ Reference list (including ERA5 and ONI)
- ☐ Word count: _____ (up to ~2,200 is fine at this stage)
- ☐ Saved as a dated version

IT NEEDS TO BE COMPLETE, NOT GOOD

Next week is the **‘hit-by-the-bus’ draft** (Schultz §13.1): complete enough that someone else could submit it. Polishing starts after that.

NOTEBOOK STRAND AND READING (BEFORE 3 DEC)

Notebook: freeze the analysis; polish figures only. Check every number in the text matches the figures.

Read: Schultz §4.9–4.11 (Results, Discussion, Conclusion), §7.2 (limitations).

Draft complete: from writer to editor

Goal: Leave with a complete draft, honest feedback on its structure, and a plan for the next nine weeks.

Reverse outline of my own draft

Mark: × = two jobs in one paragraph; → = wrong section; ? = missing step.

¶	Job	Point in ≤ 6 words	× → ?
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			

Peer-read form: the four propositions (reviewer completes)

Structure only today: no grammar or spelling comments. Reviewer: _____

	Question	Answer
Importance	After the Introduction, write the author's question in your own words.	

	Question	Answer
Success	Does the Conclusion answer that question? Quote the sentence that does.	
Value	Which figure or paragraph could go? (Or why could none?)	
Competence	One place where you doubted a method, a number or a claim.	
What works	The strongest part of this draft, and why.	

Robustness checks (choose one or two)

Check	Done?	Result in one sentence
Threshold $\pm 0.5 \rightarrow \pm 1.0$ °C	☐	
Shift region box by 5°	☐	
Leave out 2015–16	☐	
Remove linear trend	☐	
Significance with n_{eff} , or shift test (methods_demos §7)	☐	

Reproducible? Record VARIABLE, ENSO_THRESH, CLIM_START/END and region bounds for every figure:

My personal list of weaknesses

Schultz keeps a list of his own habitual weaknesses and searches every manuscript for them before submitting (§13.10). Start yours from today's feedback.

	Weakness (e.g. 'I write "it can be seen that"', 'topic sentences describe the figure')	How I'll find it (search term / check)
1		
2		

	Weakness (e.g. 'I write "it can be seen that"', 'topic sentences describe the figure')	How I'll find it (search term / check)
3		
4		
5		
6		

My revision plan to 5 February

Pass	Focus	My target date	Done
Rest	Leave the draft alone for at least a week		☐
1 Structure	Act on the reverse outline and peer read: sections and paragraphs		☐
2 CPR	Concision, precision, revision of links; get to $\leq 2,000$ words		☐
3 Final checks	Final checklist (next page), on paper; read aloud once		☐
Submit	Friday 5 February: method confirmed by your tutor	5 Feb 2027	☐

Remember AI Minimal applies to every pass: grammar and spelling support only, on text you wrote. Keep dated versions.

Final checklist before you submit

- ☐ The question is stated in the **Introduction** and answered in **sentence 1 of the Conclusion**
- ☐ The Introduction moves context → problem → purpose; no resounding banality in sentence 1
- ☐ Methods could be **repeated** by a classmate; no code; no interpretation
- ☐ Every Results paragraph starts with a **finding**; every finding is **quantified**
- ☐ Every number has **units**, a **baseline**, and a **period or sample size**; percentage points used for changes in cloud cover
- ☐ No 'significant' unless a test was done; correlation not written as causation
- ☐ Discussion gives a **physical mechanism**, honest **limitations** and returns to the question; speculation flagged
- ☐ Conclusion has **no new material** and ends with something specific
- ☐ Every figure is **cited in order**, with a complete caption and **attribution**
- ☐ Every citation is in the reference list, and vice versa; **ERA5** and **ONI** cited
- ☐ **Consistent** tag-phrases; abbreviations defined at first use
- ☐ **≤ 2,000 words** (check what counts with your tutor)
- ☐ Spelling and grammar checked; **personal weakness list** checked
- ☐ Title is informative and **accurate** (doesn't promise more than you show)

SCHULTZ'S STOPPING RULE (§13.10)

"If you average one minor revision per page or less, then the manuscript is probably ready to submit."

Toolkit: phrases, words and references

Linking words that say what the relationship is (Schultz §8.2.3)

Relationship	Examples
Contrast	however · whereas · in contrast · conversely · although
Cause and effect	therefore · consequently · because · hence · as a result
Example / detail	for example · specifically · in particular · such as
Addition / sequence	furthermore · also · first, second (not 'firstly') · then
Conclusion	in summary · overall · thus

Titles: five tests (Schultz §3.1)

Informative · accurate · clear · concise · attention-commanding. Avoid 'An investigation into...'. Don't promise more than you show ('causes' for a correlation). Write three versions (plain, question, finding) and keep the most accurate.

Precision checks for January

- **Voice:** first person is fine used sparingly; don't write all of Methods as 'I did... then I...'
- **Tense:** past for what you did and found ('was computed', 'was lower'); present for established facts ('ENSO is...').
- **Terms:** one name per thing; abbreviations defined once.
- **Units:** a space between number and unit (0.5 °C); **percentage points** for changes in cloud cover.
- **Figures:** every figure cited, in order; numbers match between caption and text.
- **Words:** 'use' not 'utilise'; 'method' not 'methodology'.

Useful Discussion phrases

- This is consistent with ...
- One explanation is ...; an alternative is ...
- This differs from [source], possibly because ...
- We speculate that ...
- This result does not hold for ..., which suggests ...

Words and phrases to cut (after Schultz Tables 10.1-10.2)

Instead of...	Write...
it can be seen that · it should be noted that · the results show that	(delete)
in order to	to
due to the fact that	because
a greater number of	more
in the vicinity of	near
is shown to be · was found to be	is · was

Instead of...	Write...
utilise · methodology	use · method
Figure 3 shows that X increases	X increases (Fig. 3)
(time) evolution · period (of time) · smaller (in size)	evolution · period · smaller

Reading from Schultz (2009) by week

Before	Read
15 Oct	§4.1–4.2; ch. 7
29 Oct	§3.1–3.2; §4.6–4.7; §12.1–12.3
12 Nov	ch. 6; §4.8; §18.3
19 Nov	§10.1; §11.10–11.14
26 Nov	ch. 8; §7.3; §9.1
3 Dec	§4.9–4.11; §7.2
January	ch. 13 (editing and finishing up)

References

Hersbach, H., and co-authors (2020) The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146, 1999–2049.

NOAA Climate Prediction Center. Oceanic Niño Index (ONI). Cite the page and the date you accessed it.

Schultz, D. M. (2009) *Eloquent Science: A Practical Guide to Becoming a Better Writer, Speaker, and Atmospheric Scientist*. American Meteorological Society.

Aldridge, J. and Derrington, A. M. *The Research Funding Toolkit*. SAGE (source of the four propositions).